Digital-Twin-Assisted Graph Learning for Joint Detection and Localization of Stealthy AC False Data Injection Attacks in Smart Grids

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***Abstract*—Stealthy false data injection attacks (FDIAs) shift the estimated state of a power system without increasing the state-estimation residual, so the conventional χ^2 bad-data detector cannot see them. This paper proposes DT-GCN, a framework that combines a physics-based digital twin (DT) with a graph convolutional network to detect and localize such attacks. The twin runs AC power flow on very-short-term injection forecasts and schedules, which gives an operating-state reference that does not use the possibly compromised telemetry. Bus-level twin-residual features are processed on the grid topology by a dual-head GCN that outputs an attack probability for the snapshot and for each bus. We evaluate the method on the IEEE 14- and 118-bus systems using AC power flow, nonlinear weighted-least-squares estimation, 15-min load and photovoltaic profiles, and AC-stealthy attacks. The χ^2 test detects only 0.89% of these attacks. On the 118-bus system, DT-GCN reaches 97.45% detection F1 at 1.35% false-positive rate, against at most 92.66% for SVM, random forest, MLP and GCN detectors trained on raw telemetry. It identifies the exact set of compromised buses in 50.78% of attacks, with 79.35% bus-level F1. Inference takes under 0.1 ms per snapshot, and the code is released for reproducibility.**

***Index Terms*—Cybersecurity, digital twin, false data injection attack, graph neural network, smart grid, state estimation, attack localization.**

I. Introduction

Modern power systems depend on a cyber layer of remote terminal units, phasor measurement units, and wide-area networks. That layer feeds the energy management system (EMS). Weighted-least-squares (WLS) state estimation (SE) turns raw telemetry into the operating state used by contingency analysis, optimal power flow and markets [1]. The classical bad-data detector (BDD) tests the SE residual with a χ^2 test. Liu *et al.* [2] showed that an adversary who knows the network model can craft a *stealthy* false data injection attack (FDIA). Such an attack moves the estimated state while leaving the residual unchanged. Hug and Giampapa [3] extended this to the full nonlinear AC model. The 2015 Ukrainian incident made coordinated cyber intrusions into grid control systems a practical concern [4].

Data-driven detectors, including support vector machines [5], [6], deep networks [7], and graph neural networks (GNNs) [8], [9], can flag stealthy FDIAs. Most of them learn directly from raw telemetry, where the attack signature is buried under large load and renewable variability [10]. Meanwhile, the *digital twin* (DT) has become a central concept for next-generation control centers [11], [12]. A DT is a continuously synchronized, physics-based replica of the grid. It can supply an *independent* expectation of the operating state from forecasts and schedules that never pass through the compromised telemetry channel. This is related in spirit to forecasting-aided SE [13].

This paper combines the two ideas. Its contributions are:

1) A DT-assisted detection architecture. A physics-based twin runs AC power flow on very-short-term injection forecasts. Its outputs are converted into six physically meaningful, bus-level *twin-residual* features that whiten normal operating variability.
2) A dual-head graph convolutional network (DT-GCN) that consumes these features on the grid topology. It jointly *detects* an attack (graph level) and *localizes* the compromised buses (node level).
3) A reproducible open-source benchmark on the IEEE 14- and 118-bus systems. It includes AC power flow, a nonlinear WLS estimator, realistic 15-min load/PV profiles, and AC-stealthy attacks. The benchmark compares eight detectors, ablations, and sensitivity to twin fidelity.

II. System Model

A. AC Measurement Model and State Estimation

For an n-bus, m-branch network with state $\mathbf{x} = [\boldsymbol{\theta}^\top, \mathbf{V}^\top]^\top$ (the reference angle is excluded), the telemetry is

$$\mathbf{z} = \mathbf{h}(\mathbf{x}) + \mathbf{e}, \quad \mathbf{e} \sim \mathcal{N}(\mathbf{0}, \mathbf{R}), \tag{1}$$

where $\mathbf{z} \in \mathbb{R}^{3n+2m}$ stacks bus voltage magnitudes, active/reactive injections and from-end active/reactive branch flows. $\mathbf{h}(\cdot)$ is built from the bus and branch admittance matrices. WLS SE solves $\hat{\mathbf{x}} = \arg\min_{\mathbf{x}} J(\mathbf{x}) = (\mathbf{z} - \mathbf{h}(\mathbf{x}))^\top \mathbf{R}^{-1} (\mathbf{z} - \mathbf{h}(\mathbf{x}))$ by Gauss–Newton iterations with the analytic Jacobian $\mathbf{H}(\mathbf{x})$. The BDD raises an alarm when $J(\hat{\mathbf{x}}) > \chi^2_{p-q,\,0.99}$, where p and $q = 2n - 1$ are the numbers of measurements and states.

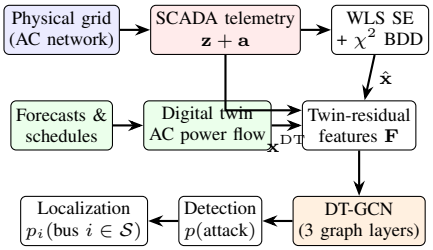


Fig. 1. DT-assisted detection and localization pipeline. The twin path (green) never consumes field telemetry, so it is unaffected by the injection.

B. AC-Stealthy FDIA

An attacker with model knowledge and an estimate $\hat{\mathbf{x}}$ selects a connected set $\mathcal{S}$ of $k \in \{1, 2, 3\}$ non-reference buses and a state bias $\mathbf{c}$ supported on $\mathcal{S}$. It then injects

$$\mathbf{a} = \mathbf{h}(\hat{\mathbf{x}} + \mathbf{c}) - \mathbf{h}(\hat{\mathbf{x}}). \tag{2}$$

Because $\mathbf{z}+\mathbf{a}$ is consistent with $\hat{\mathbf{x}}+\mathbf{c}$, the SE converges to the biased state and J is unchanged, so the attack is invisible to the BDD [2], [3]. Only measurements incident to $\mathcal{S}$ are modified. Angle biases are drawn as $|c_\theta| \sim \mathcal{U}(0.001, 0.02)$ rad (0.06°–1.15°) with a common sign. In half of the attacks a voltage bias $|c_V| = (0.2\text{–}0.5)|c_\theta|$ is added. For completeness, *naive* attacks that add 5–15σ biases to 3–7 random meters are also included.

III. Proposed DT-GCN Framework

A. Physics-Based Digital Twin

Fig. 1 shows the architecture. At each 15-min SE cycle the twin receives very-short-term bus-level injection forecasts $\tilde{P}_{d,i} = P_{d,i}(1 + \epsilon_i)$, $\epsilon_i \sim \mathcal{N}(0, \sigma_f^2)$, and the generation schedule. It then solves a full Newton–Raphson AC power flow to obtain $\mathbf{x}^{\mathrm{DT}}$ and the expected telemetry $\mathbf{z}^{\mathrm{DT}} = \mathbf{h}(\mathbf{x}^{\mathrm{DT}})$. The default fidelity is $\sigma_f = 2\%$, which is typical of 15-min-ahead nodal forecasts; Section V-D varies it from 1% to 8%. Because the twin is driven by forecasts and dispatch rather than field meters, an injection that biases $\hat{\mathbf{x}}$ by $\mathbf{c}$ produces a discrepancy $\hat{\mathbf{x}} - \mathbf{x}^{\mathrm{DT}} \approx \mathbf{c} + \boldsymbol{\delta}_f$, where $\boldsymbol{\delta}_f$ is the small twin error.

B. Twin-Residual Graph Features

For every bus i we form the feature vector $\mathbf{f}_i \in \mathbb{R}^6$:

$$\mathbf{f}_i = \big[\Delta\theta_i,\ \Delta\theta_i - \tfrac{1}{|\mathcal{N}_i|}\textstyle\sum_{j\in\mathcal{N}_i}\Delta\theta_j,\ \Delta V_i,\ r_i^P,\ r_i^Q,\ \textstyle\sum_{\ell\ni i}|r_\ell^F|\big], \tag{3}$$

where $\Delta(\cdot) = \hat{(\cdot)} - (\cdot)^{\mathrm{DT}}$, r_i^P, r_i^Q are injection residuals $\mathbf{z} - \mathbf{z}^{\mathrm{DT}}$, and r_ℓ^F are flow residuals on incident branches. The neighbour-relative angle term cancels the angle drift that accumulates away from the slack bus. Each feature is z-scored per bus using *attack-free* training samples only.

C. Dual-Head Graph Convolutional Network

Let $\hat{\mathbf{A}} = \mathbf{D}^{-1/2}(\mathbf{A}+\mathbf{I})\mathbf{D}^{-1/2}$ be the normalized adjacency of the network graph [14]. Each of the $L = 3$ layers computes

$$\mathbf{H}^{(l+1)} = \mathrm{GELU}\big(\hat{\mathbf{A}}\mathbf{H}^{(l)}\mathbf{W}^{(l)} + \mathbf{H}^{(l)}\mathbf{U}^{(l)}\big), \tag{4}$$

with 64 hidden units, a residual projection $\mathbf{U}^{(l)}$ and dropout 0.1. A node head (linear + sigmoid) outputs p_i, the probability that bus $i \in \mathcal{S}$. A graph head (MLP on concatenated mean- and max-pooled embeddings) outputs the detection probability p_g. The loss is $\mathcal{L} = \mathrm{BCE}(p_g, y) + \lambda\,\mathrm{BCE}_w(\mathbf{p}, \mathbf{y}^{\mathrm{node}})$ with $\lambda = 1$ and a positive-class weight that compensates for the sparsity of compromised buses. Training uses AdamW (learning rate 2×10^{-3}, weight decay 10^{-4}, cosine schedule, 60 epochs, batch 128), and the epoch with the lowest validation loss is kept.

IV. Experimental Setup

A. Environment and Data

All code is written in Python using pandapower [15], which supplies the MATPOWER-format [16] test cases and admittance builders. Detectors use PyTorch [17] and scikit-learn [18]. Every experiment runs on a 4-core CPU, and the full pipeline is reproduced by one script. For each of the IEEE 14- and 118-bus systems, 6000 consecutive 15-min snapshots (62.5 days) are generated. The system load has a double-peak daily shape, a weekend reduction and AR(1) drift. Bus-level loads carry an extra AR(1) variation with 3% standard deviation. Photovoltaic plants with clear-sky profiles and day-to-day cloudiness sit at 30% of load buses (25% of peak load). Generators follow net load, and the slack bus balances losses. Measurement noise standard deviations are 0.004 pu (voltage), 0.01 pu (injections) and 0.008 pu (flows) on a 100 MVA base. Each snapshot is attack-free with probability 0.4, carries a stealthy FDIA with probability 0.5, or carries a naive attack with probability 0.1. The data are split *chronologically* into 70% training, 15% validation and 15% test, so test days are never seen during training.

B. Compared Detectors

We compare: (i) the χ^2 BDD; (ii) *DT-Threshold*, a learning-free rule that flags the largest standardized twin residual with its threshold set at 1% false-positive rate (FPR) on validation; (iii) SVM-RBF [5] and (iv) random forest (RF, 300 trees) on standardized raw telemetry; (v) MLP and (vi) GCN on raw per-bus telemetry, representative of [7] and [9]; (vii) DT-MLP, an ablation without the graph; and (viii) the proposed DT-GCN. Models (v)–(viii) share the same dual-head loss. Learning-based results are mean±std over five seeds (SVM: one seed, because it is deterministic). Localization is scored on stealthy test samples with micro-averaged bus-level precision, recall and F1, and with exact-match accuracy, meaning the predicted bus set equals $\mathcal{S}$.

TABLE I

ATTACK DETECTION ON THE CHRONOLOGICAL TEST SET (%, MEAN±STD OVER FIVE SEEDS; BEST IN BOLD). THE TEST SETS CONTAIN 347/453/100 (14-BUS) AND 370/451/79 (118-BUS) NORMAL/STEALTHY/NAIVE SNAPSHOTS.

	IEEE 14-bus				IEEE 118-bus			
Method	Acc.	F1	FPR ↓	AUC	Acc.	F1	FPR ↓	AUC
BDD (chi-square)	49.56	31.00	**0.86**	58.54	49.67	26.10	0.81	57.79
DT-Threshold	95.67	96.39	1.73	98.34	96.00	96.50	**0.54**	98.74
SVM-RBF (raw)	94.78	95.62	2.02	97.71	85.78	87.33	10.54	92.58
RandomForest (raw)	93.24±0.12	94.42±0.11	6.34±0.29	97.38±0.04	81.27±0.80	83.61±0.61	18.49±1.85	89.51±0.37
MLP (raw)	93.89±0.46	94.94±0.36	5.19±1.48	97.65±0.14	78.62±1.34	81.19±0.89	20.92±5.16	87.27±1.10
GCN (raw)	93.09±0.14	94.13±0.14	2.42±0.85	96.81±0.15	91.84±0.10	92.66±0.10	1.84±0.59	94.86±0.22
DT-MLP	95.58±0.43	96.36±0.34	3.75±1.37	98.61±0.07	85.29±0.90	87.05±0.78	12.86±4.84	92.96±0.37
DT-GCN (prop.)	**97.27±0.24**	**97.75±0.19**	1.96±0.69	**99.23±0.06**	**97.04±0.36**	**97.45±0.31**	1.35±0.43	**99.00±0.05**

TABLE II

LOCALIZATION OF COMPROMISED BUSES ON STEALTHY TEST ATTACKS (%, MEAN±STD OVER FIVE SEEDS).

	IEEE 14-bus			IEEE 118-bus		
Method	Recall	F1	Exact match	Recall	F1	Exact match
DT-Threshold	82.34	52.63	2.65	84.02	49.13	2.88
MLP (raw)	90.99±0.54	79.21±2.02	34.97±7.32	43.20±3.67	32.58±1.30	0.62±0.62
GCN (raw)	87.61±0.41	60.22±0.97	13.95±1.89	84.18±0.50	39.71±1.03	2.00±0.41
DT-MLP	94.13±0.33	**89.26±1.80**	66.18±6.38	74.48±2.96	45.58±0.98	1.86±0.60
DT-GCN (prop.)	**95.93±0.20**	89.22±0.92	**77.53±1.78**	**94.46±0.66**	**79.35±3.55**	**50.78±5.92**

V. RESULTS AND DISCUSSION

A. Detection Performance

Table I and Fig. 2 summarize detection. The χ^2 BDD catches 100.00% and 96.20% of naive attacks. It detects only 0.44% and 0.89% of stealthy FDIAs, which is no better than its 1% nominal false-alarm rate. This confirms that the generated attacks are AC-stealthy. On the 14-bus system all learning-based detectors exceed 94% F1. Because the model is small, raw-telemetry classifiers can still learn the narrow normal range of each bus injection. The proposed DT-GCN reaches the highest F1 (97.75%) and AUC (99.23%).

The advantage widens on the 118-bus system. There, 726 measurements and strongly varying PV and load make the raw-data decision surface much harder. SVM, RF and MLP on raw telemetry fall to 81–87% F1 with FPR above 10%. The GCN on raw telemetry reaches 92.66% F1. DT-GCN keeps 97.45% F1 at 1.35% FPR, with AUC 99.00%. Fig. 2 shows that DT-GCN holds about 95% true-positive rate even at 10^{-3} FPR, which matters for operators who must keep alarm fatigue low.

The ablations separate the two ingredients. The learning-free DT-Threshold is already strong (F1 96.50% on 118-bus), so the twin residuals themselves carry most of the detection signal. Removing the graph (DT-MLP) *hurts* on the large system (87.05% F1). A flattened 708-dimensional input overfits the 4200 training snapshots, while the topology-shared GCN weights scale to large networks.

B. Sensitivity to Attack Strength and Size

Fig. 3 breaks the detection rate down by the injected angle bias. The twin-based detectors find essentially all attacks above about $0.3°$. The raw-telemetry RF still misses 6–14% of them on the 118-bus system. In the hardest bin ($0.06°$–$0.11°$, comparable to the twin's own angle error) no detector exceeds 56%, which marks the detectability floor set by twin fidelity and noise. The proposed method gains most between $0.11°$ and $0.23°$ on the 118-bus system. There, DT-GCN detects 88% and 100% of attacks in the two bins, while the best competitor detects 76% and 82%. On the 14-bus system, DT-GCN and DT-MLP perform similarly in this range (87–90%). For attacks spanning $k = 1, 2, 3$ buses on the 118-bus system, DT-GCN detects 98.0/96.8/96.5%, compared with 78.0/90.5/95.1% for RF on raw telemetry. Single-bus attacks are therefore not harder for the proposed method, whereas raw-data methods lose up to 17 percentage points on them.

C. Localization

Table II reports bus-level localization, where DT-GCN gives the clearest gain. On the 118-bus system it achieves 79.35% F1 and identifies the *exact* compromised bus set in 50.78% of attacks. The best alternative reaches 49.13% F1 and below 3% exact match. The threshold detector has high recall but poor precision: an attack on bus i also perturbs residuals at its neighbors. Message passing over the topology lets the GCN learn which bus of a disturbed neighborhood is the *source*. On the 14-bus system, DT-GCN and DT-MLP have similar F1 (89.22% vs 89.26%), but DT-GCN's exact-match rate is 11 points higher (77.53%). Fig. 5 shows a correctly localized three-bus attack (buses 5, 6 and 13).

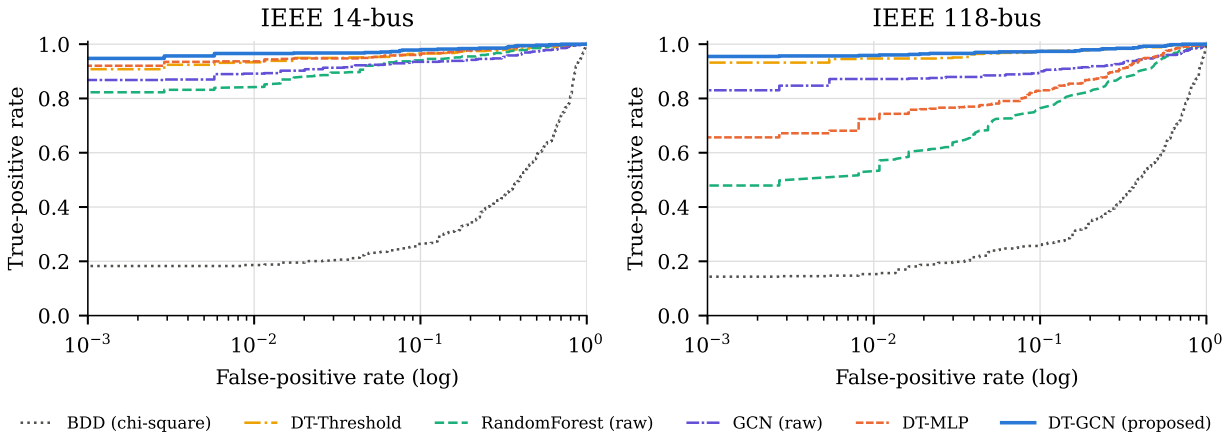


Fig. 2. ROC curves (log-scaled FPR) on the test set.

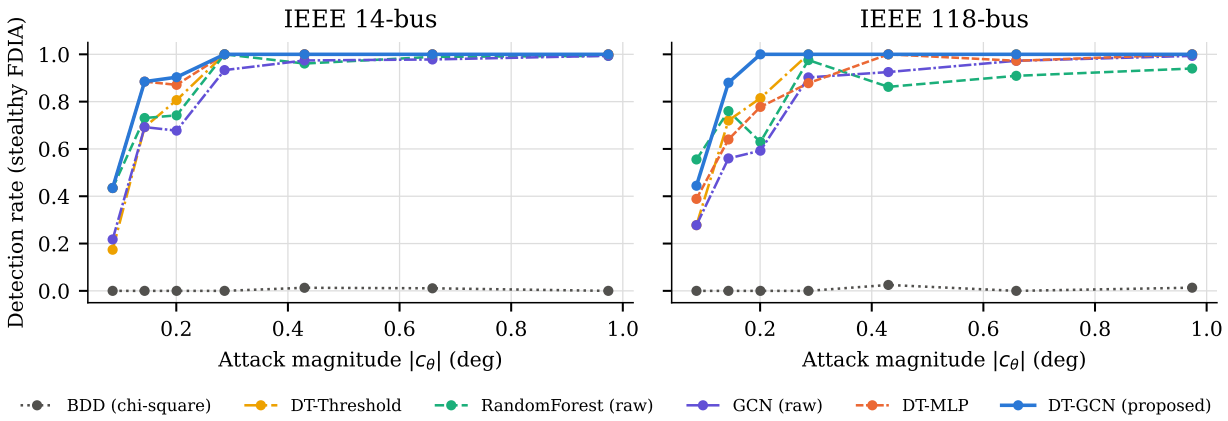


Fig. 3. Detection rate of stealthy FDIAs versus angle-bias magnitude.

D. Sensitivity to Digital-Twin Fidelity

The method depends on the twin being accurate. Fig. 4 retrains the DT-based detectors with the twin's forecast error swept from 1% to 8%. For each level the twin is re-simulated with a new forecast-error realization. Detection degrades gracefully. On the 14-bus system DT-GCN keeps 97.9%, 96.7% and 95.7% F1 at $\sigma_f = 1$, 4 and 8%, so even at 8% it matches the best raw-telemetry detector (SVM, 95.6%). On the 118-bus system it drops from 98.3% to 94.4% F1, compared with 97.9% to 90.9% for DT-Threshold and 88.3% to 79.7% for DT-MLP. The graph model is therefore the most robust *detector* when the twin is imperfect. Localization is more sensitive. On the 118-bus system, DT-GCN's bus-level F1 falls from 89.5% ($\sigma_f = 1\%$) to 64.6% (4%) and 54.4% (8%), approaching the twin-threshold level of about 50%. On the 14-bus system the decline is milder (90.8% to 77.1%), and there DT-MLP degrades less (84.0% at 8%). Accurate localization in large networks therefore requires a well-calibrated twin, with nodal forecast errors of a few percent. This is a practical design target for DT-enabled control centers.

E. Computational Cost and Deployment

The twin power flow, WLS estimation and feature extraction take about 30 ms per snapshot on the 118-bus system in our pure-Python implementation. Batched DT-GCN inference takes 0.078 ms per snapshot. The whole chain is therefore far inside a 15-min (or even a 1-s PMU) SE cycle. The model has 25.7k parameters and runs on a commodity CPU. In practice it would run in parallel with the existing BDD, as a second line of defense that needs no change to the SE. The bus-level output can trigger targeted measures, such as re-authenticating the remote terminal units at the flagged buses or temporarily replacing their measurements with twin pseudo-measurements.

F. Limitations

This study uses simulated data on standard test systems. Load, PV and forecast errors follow parametric models. The attacker is assumed not to compromise the forecast and scheduling channels that feed the twin; an adversary who can also bias the twin inputs is outside this threat model. Topology changes and parameter errors in the twin were not modeled. Detectors were trained with labeled attacks, so

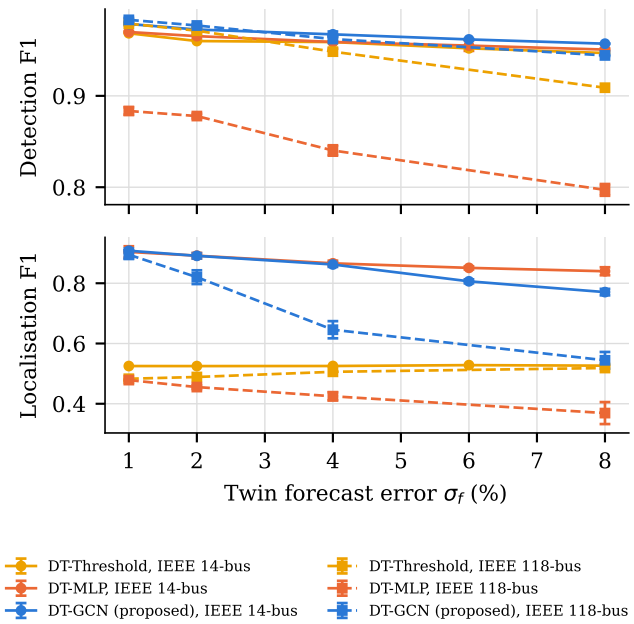


Fig. 4. Effect of twin forecast error σ_f on detection and localization (solid: 14-bus, 3 seeds; dashed: 118-bus, 2 seeds).

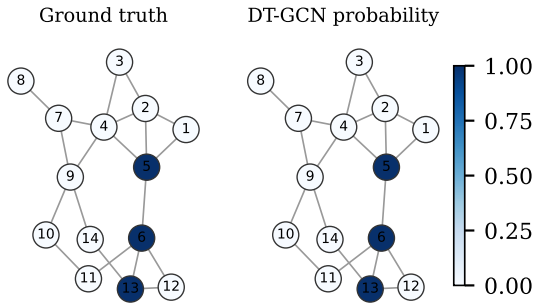


Fig. 5. Localization of a three-bus stealthy FDIA on the IEEE 14-bus system.

robustness to attack strategies unseen in training should be checked with semi-supervised or one-class variants. These are planned extensions toward field validation on utility data.

VI. Conclusion

This paper presented DT-GCN, a digital-twin-assisted graph learning framework for detecting and localizing stealthy AC false data injection attacks. A forecast-driven physics twin supplies an attack-independent reference state. Its bus-level residuals turn a hard anomaly-detection problem in raw telemetry into a well-conditioned graph problem. On the IEEE 14- and 118-bus systems with realistic load/PV variability, a nonlinear WLS estimator and AC-stealthy attacks invisible to the χ^2 test, DT-GCN achieved 97.75% and 97.45% detection F1. On the 118-bus system it localized the exact compromised buses in 50.78% of attacks, outperforming SVM, random forest, MLP and GCN detectors on raw telemetry and the learning-free twin threshold. Future work will cover twin-input integrity,

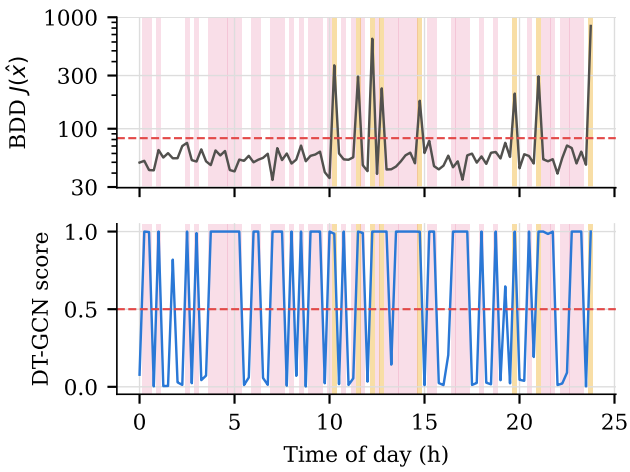


Fig. 6. One test day on the IEEE 14-bus system. Pink: stealthy FDIA intervals; amber: naive attacks; dashed: alarm thresholds. The BDD (top) responds only to naive attacks. DT-GCN (bottom) flags both.

topology attacks, PMU-rate operation, and hardware-in-the-loop validation.

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